LEB-DS

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[17BS5CS01]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2019

INFERENTIAL STATISTICS

Time:3 Hours **Maximum Marks:70**

Instructions: i. Answer any **FIVE** full Questions choosing at least two from each part.

ii. Missing data may be suitably assumed.

PART-A

1. a. Explain about statistical hypothesis and types of error with respect to false positive and false negative. **08**

b. Suppose $X_1, X_2, \ldots, X_n$ is a random sample from an exponential distribution with parameter θ. Are the hypotheses $H : \theta = 3$ and $H : \theta > 3$ a simple or a composite hypothesis? And if it were a normal distribution with mean μ and unknown variance σ^2, would the hypothesis $H : \mu = 12$ a simple or a composite hypothesis? **06**

2. a. Give the definition for *most powerful test*, and *uniformly most powerful test*. Suppose X is a single observation (again, one data point!) from a population with probability density function given by: **07**

$$f(x) = \theta x^{\theta-1}$$

for $0 < x < 1$. Find the test with the best critical region, that is, find the most powerful test, with significance level $\alpha = 0.05$, for testing the simple null hypothesis $H_0 : \theta = 3$ against the simple alternative hypothesis $H_A : \theta = 2$.

b. Suppose $X_1, X_2, \ldots, X_n$ is a random sample from a normal population with mean μ and variance 16. Find the test with the best critical region, that is, find the most powerful test, with a sample size of $n = 16$ and a significance level $\alpha = 0.05$ to test the simple null hypothesis $H_0: \mu = 10$ against the simple alternative hypothesis $H_A: \mu = 15$. **07**

3. a. i. A coin is tossed 1050 times and lands on heads 500 times. Construct a 90% confidence interval for the probability p of getting a head. **07**

ii. An automobile company is looking for fuel additives that might increase gas mileage. Without additives, their cars are known to average 25 mpg (miles per gallons) with a standard deviation of 2.4 mpg on a road trip from London to Edinburgh. The company now asks whether a particular new additive increases this value. In a study, thirty cars are sent on a road trip from London to Edinburgh. Suppose it turns out that the thirty cars averaged $\bar{x} = 25.5$ mpg with the additive. Can we conclude from this result that the additive is effective?

b. i. A supermarket has come under scrutiny after a number of complaints that its carrier bags fall apart when the load they carry is 5 kg. Out of a random sample of 200 bags, 185 do not tear when carrying a load of 5kg. Can the supermarket claim at a 5% significance level that more that 90% of the bags will not fall apart? **07**

ii. A motor manufacturer wishes to replace steel suspension components by aluminium components to save weight and thereby improve performance and fuel consumption. Tensile strength tests are carried out on randomly chosen samples of two possible components before a final choice is made. The results are:

Component Number	Sample Size	Mean Tensile Strength (kg mm^{-2})	Standard Deviation (kg mm^{-2})
1	15	90	2.3
2	10	88	2.2

Is there any difference between the measured tensile strengths at the 5% level of significance?

4. a. A randomized trial is designed to evaluate the effectiveness of a newly developed pain reliever designed to reduce pain in patients following joint replacement surgery. The trial compares the new pain reliever to the pain reliever currently in use (called the standard of care). A total of 100 patients undergoing joint replacement surgery agreed to participate in the trial. Patients were randomly assigned to receive either the new pain reliever or the standard pain reliever following surgery and were blind to the treatment assignment. Before receiving the assigned treatment, patients were asked to rate their pain on a scale of 0-10 with higher scores indicative of more pain. Each patient was then given the assigned treatment and after 30 minutes was again asked to rate their pain on the same scale. The primary outcome was a reduction in pain of 3 or more scale points (defined by clinicians as a clinically meaningful reduction). The following data were observed in the trial. **07**

Treatment Group	n	Number with Reduction of 3+ Points	Proportion with Reduction of 3+ Points
New Pain Reliever	50	23	0.46
Standard Pain Reliever	50	11	0.22

Test whether there was a significant difference in the proportions of patients reporting a meaningful reduction (i.e., a reduction of 3 or more scale points) using Chi-square test (Critical value = 3.84).

b. A personal trainer is interested in comparing the anaerobic thresholds of elite athletes. Anaerobic threshold is defined as the point at which the muscles cannot get more oxygen to sustain activity or the upper limit of aerobic exercise. It is a measure also related to maximum heart rate. The following data are anaerobic thresholds for distance runners, distance cyclists, distance swimmers and cross-country skiers. **07**

Distance Runners	Distance Cyclists	Distance Swimmers	Cross-Country Skiers
185	190	166	201
179	209	159	195
192	182	170	180
165	178	183	187
174	181	160	215

Applying The Kruskal-Wallis Test, state if there is a difference in anaerobic thresholds among the different groups of elite athletes (The critical value is 7.81).

PART - B

5. a. Explain the Central Limit Theorem with real life example. Why is it important? 06

b. i. Explain Basu's Theorem and its Role in Statistical Inference. 05

ii. Explain Rao-Blackwell Theorem with example. 03

6. a. For the following random samples, find the maximum likelihood function: 08

i. $X_i \sim Binomial(3,\theta)$, and we have observed $(x_1,x_2,x_3,x_4)=(1,3,2,2)$.

ii. $Xi \sim Exponential(\theta)$ and we have observed $(x_1,x_2,x_3,x_4) = (1.23,3.32,1.98,2.12)$.

b. Find the model parameters for model estimation $\hat{y} = ax + b$ using ordinary least square method. 06

7. a. Let $X_1, X_2, \ldots, X_n$ be normal random variables with mean μ and variance σ^2. What are the method of moments estimators of the mean μ and variance σ^2? 07

b. Let $X_1, X_2, \ldots, X_n$ be gamma random variables with parameters α and θ, so that the probability density function is: 07

$$f(x_i) = \frac{1}{\Gamma(\alpha)\theta^\alpha} x^{\alpha-1} e^{-x/\theta}$$

for $x > 0$. Therefore, the likelihood function:

$$L(\alpha,\theta) = \left(\frac{1}{\Gamma(\alpha)\theta^\alpha}\right)^n (x_1 x_2 \cdots x_n)^{\alpha-1} \exp\left[-\frac{1}{\theta}\sum x_i\right]$$

is difficult to differentiate because of the gamma function $\Gamma(\alpha)$. So, rather than finding the maximum likelihood estimators, what are the method of moments estimators of α and θ?

8. a. Suppose that the random variable X is transmitted over a communication channel. Assume that the received signal is given by 05

$$Y=X+W,$$

where $W \sim N(0,\sigma 2)$ is independent of X. Suppose that $X=1$ with probability p, and $X=-1$ with probability $1-p$. The goal is to decide between $X=1$ and $X=-1$ by observing the random variable Y. Find the MAP test for this problem.

b. Let $X \sim N(0,1)$ and $Y=X+W$, where $W \sim N(0,1)$ is independent of X. 09

i. Find the MMSE estimator of X given Y, $\hat{X}_M$.

ii. Find the MSE of this estimator, using $MSE=E[(X-\hat{X}_M)^2]$.

iii. Check that $E[X^2]=E[\hat{X}_M^2]+E[\tilde{X}_M^2]$.